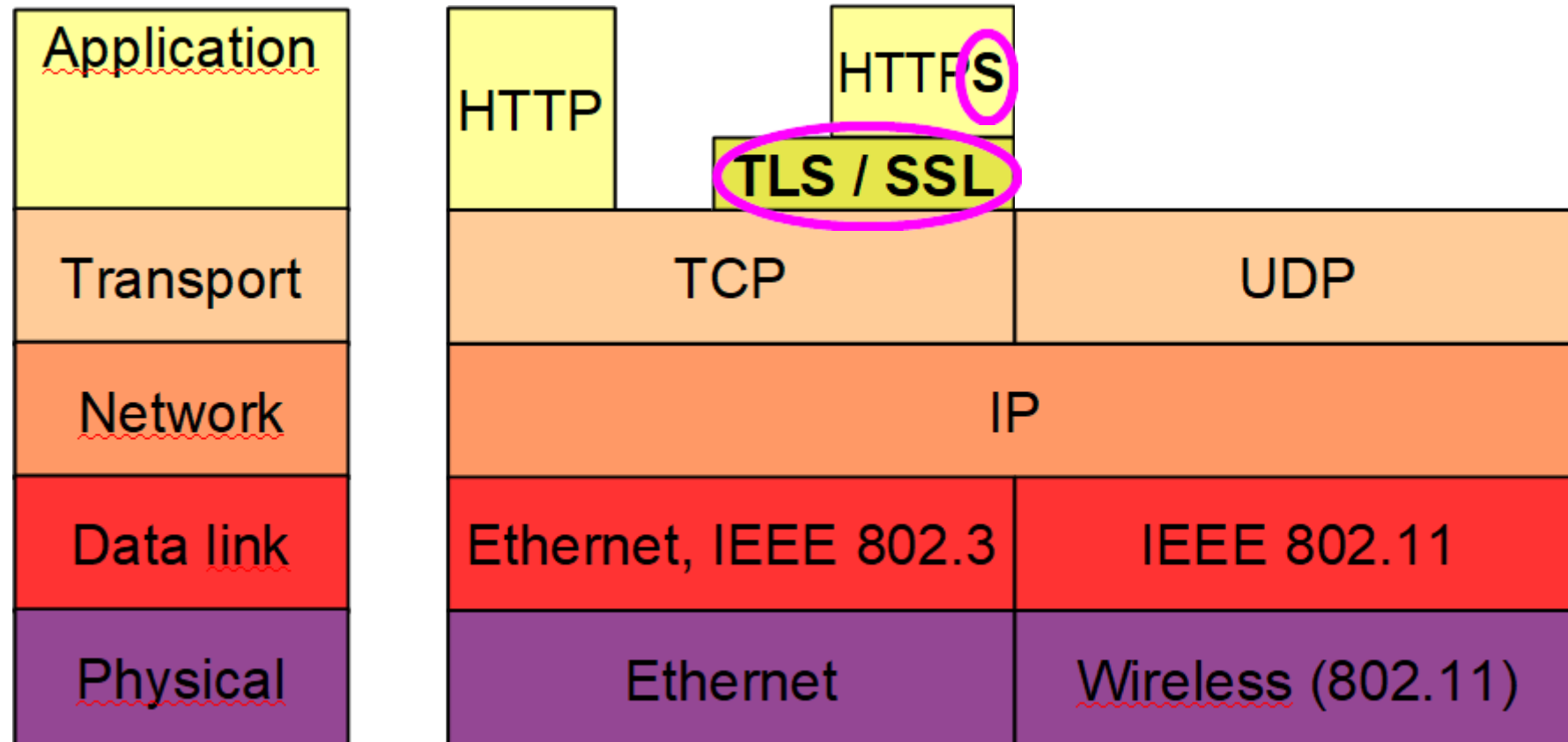


# ***Datensicherheit***

## ***TLS***

*Prof. Dr. Stefan Heiss*



- **Secure Sockets Layer Protocols**

- **SSL 1.0:** Netscape Communications Corp., 1994
- **SSL 2.0:** The SSL Protocol, Netscape Communications Corp., Feb. 1995
- **SSL 3.0:** The SSL 3.0 Protocol, Netscape Communications Corp., Nov. 1996

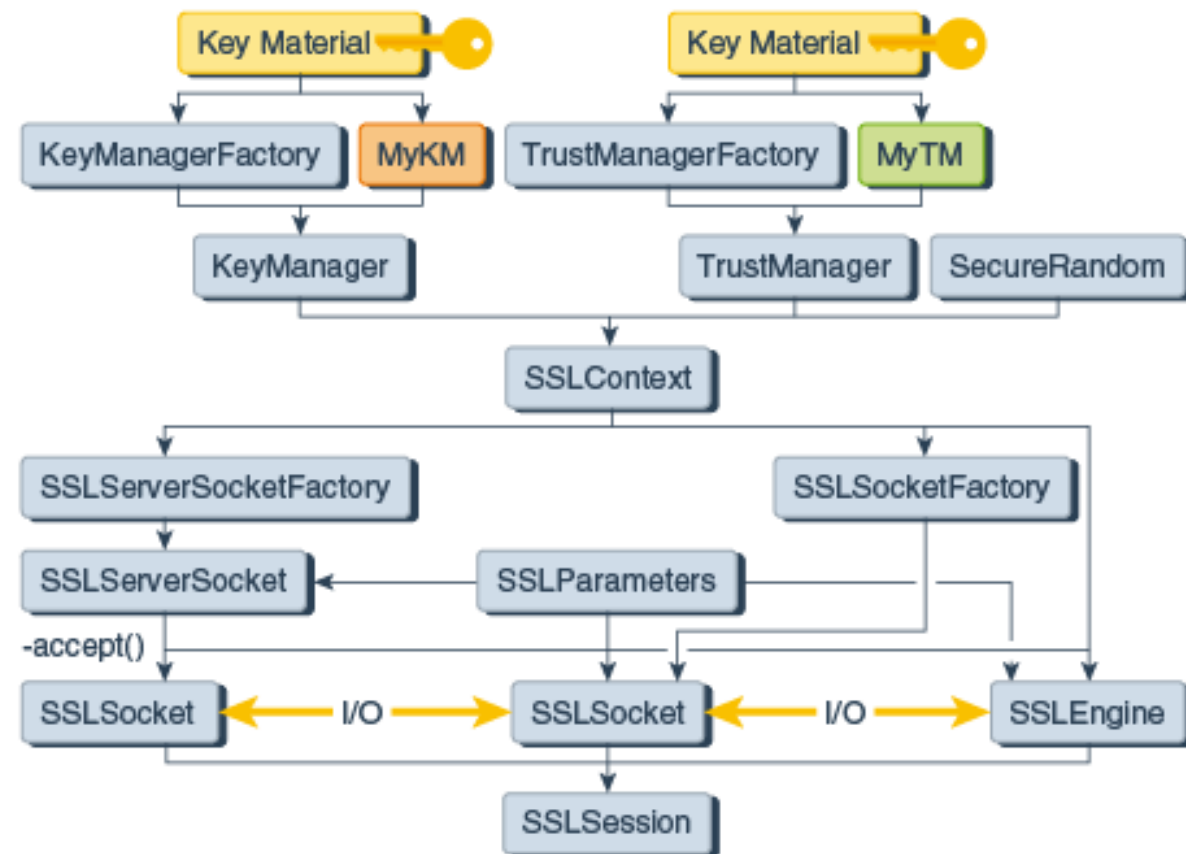
- **Transport Layer Security Protocols**

- **TLS 1.0 = SSL 3.1:** The TLS Protocol - Version 1.0, [RFC 2246](#), Jan. 1999
- **TLS 1.1:** The Transport Layer Security (TLS) Protocol - Version 1.1, [RFC 4346](#), April 2006
- **TLS 1.2:** The Transport Layer Security (TLS) Protocol - Version 1.2, [RFC 5246](#), August 2008
- **TLS 1.3:** The Transport Layer Security (TLS) Protocol - Version 1.3, [RFC 8446](#), August 2018

- Java Secure Socket Extension (JSSE)
- [JSSE Reference Guide](#) (Chapter 8 in [Security Developer's Guide](#))
- Integration von TLS 1.3 seit Java 11

TCP: Server Socket  
Socket

TLS: SSL Server Socket  
SSL Socket



- **Record Protocol**

- Privacy: symmetric encryption
- Integrity: MAC's or AEAD methods
- Keys are negotiated per connection with the help of the TLS Handshake Protocol

- **Handshake Protocol**

- Authentication of peer's identities
- Negotiation of secret session keys

- **Alert Protocol**

- **TLS v ≤ 1.2:** <Handshake Protocol Algos.> **WITH** <Record Protocol Algos.>
  - Addition of Kerberos Cipher Suites to Transport Layer Security (TLS) , [RFC 2712](#)
  - Addition of SEED Cipher Suites to Transport Layer Security (TLS), [RFC 4162](#)
  - Pre-Shared Key Ciphersuites for Transport Layer Security (TLS), [RFC 4279](#)
  - **The Transport Layer Security (TLS) Protocol - Version 1.1**, [RFC 4346](#)
  - Pre-Shared Key (PSK) Ciphersuites with NULL Encryption for Transport Layer Security (TLS), [RFC 4785](#)
  - Using the Secure Remote Password (SRP) Protocol for TLS Authentication , [RFC 5054](#)
  - **The Transport Layer Security (TLS) Protocol - Version 1.2**, [RFC 5246](#)
  - AES Galois Counter Mode (GCM) Cipher Suites for TLS, [RFC 5288](#)
  - TLS Elliptic Curve Cipher Suites with SHA-256/384 and AES Galois Counter Mode (GCM) , [RFC 5289](#)
  - DES and IDEA Cipher Suites for Transport Layer Security (TLS), [RFC 5469](#)
  - Pre-Shared Key Cipher Suites for TLS with SHA-256/384 and AES Galois Counter Mode, [RFC 5487](#)
  - ECDHE\_PSK Cipher Suites for Transport Layer Security (TLS) , [RFC 5489](#)
  - Camellia Cipher Suites for TLS, [RFC 5932](#)
  - Addition of the ARIA Cipher Suites to Transport Layer Security (TLS), [RFC 6209](#)
  - Datagram Transport Layer Security Version 1.2, [RFC 6347](#)
  - Addition of the Camellia Cipher Suites to Transport Layer Security (TLS), [RFC 6367](#)
  - AES-CCM Cipher Suites for Transport Layer Security (TLS), [RFC 6655](#)
  - AES-CCM Elliptic Curve Cryptography (ECC) Cipher Suites for TLS, [RFC 7251](#)
  - ChaCha20-Poly1305 Cipher Suites for Transport Layer Security (TLS), [RFC 7905](#)
  - Elliptic Curve Cryptography (ECC) Cipher Suites for Transport Layer Security (TLS) Versions 1.2 and Earlier, [RFC 8422](#)
  - GOST Cipher Suites for Transport Layer Security (TLS) Protocol Version 1.2, [draft-smyshlyaev-tls12-gost-suites](#)

- **TLS v = 1.3:** <Record Protocol Algos.>
  - The Transport Layer Security (TLS) Protocol Version 1.3, [RFC 8446](#)
  - Secure Password Ciphersuites for Transport Layer Security (TLS), [RFC 8492](#)
  - ShangMi (SM) Cipher Suites for TLS 1.3, [RFC 8998](#)
  - TLS 1.3 Authentication and Integrity only Cipher Suites, [draft-camwinget-tls-ts13-macciphersuites](#)
  - GOST Cipher Suites for Transport Layer Security (TLS) Protocol Version 1.3, [draft-smyshlyaev-tls13-gost-suites](#)
  
- Ciphersuite identifiers: [Transport Layer Security \(TLS\) Parameters](#)

A symmetric cipher suite defines the pair of the AEAD algorithm and hash algorithm to be used with HKDF. Cipher suite names follow the naming convention:

```
CipherSuite TLS_AEAD_HASH = VALUE;
```

| Component | Contents                                       |
|-----------|------------------------------------------------|
| TLS       | The string "TLS"                               |
| AEAD      | The AEAD algorithm used for record protection  |
| HASH      | The hash algorithm used with HKDF              |
| VALUE     | The two-byte ID assigned for this cipher suite |

← s. HMAC - Based  
Key-Derivation  
(Bsp.: TLS 1.3 -  
Schlüsselableitung)



This specification defines the following cipher suites for use with TLS 1.3.

| Description                  | Value        |
|------------------------------|--------------|
| TLS_AES_128_GCM_SHA256       | {0x13, 0x01} |
| TLS_AES_256_GCM_SHA384       | {0x13, 0x02} |
| TLS_CHACHA20_POLY1305_SHA256 | {0x13, 0x03} |
| TLS_AES_128_CCM_SHA256       | {0x13, 0x04} |
| TLS_AES_128_CCM_8_SHA256     | {0x13, 0x05} |

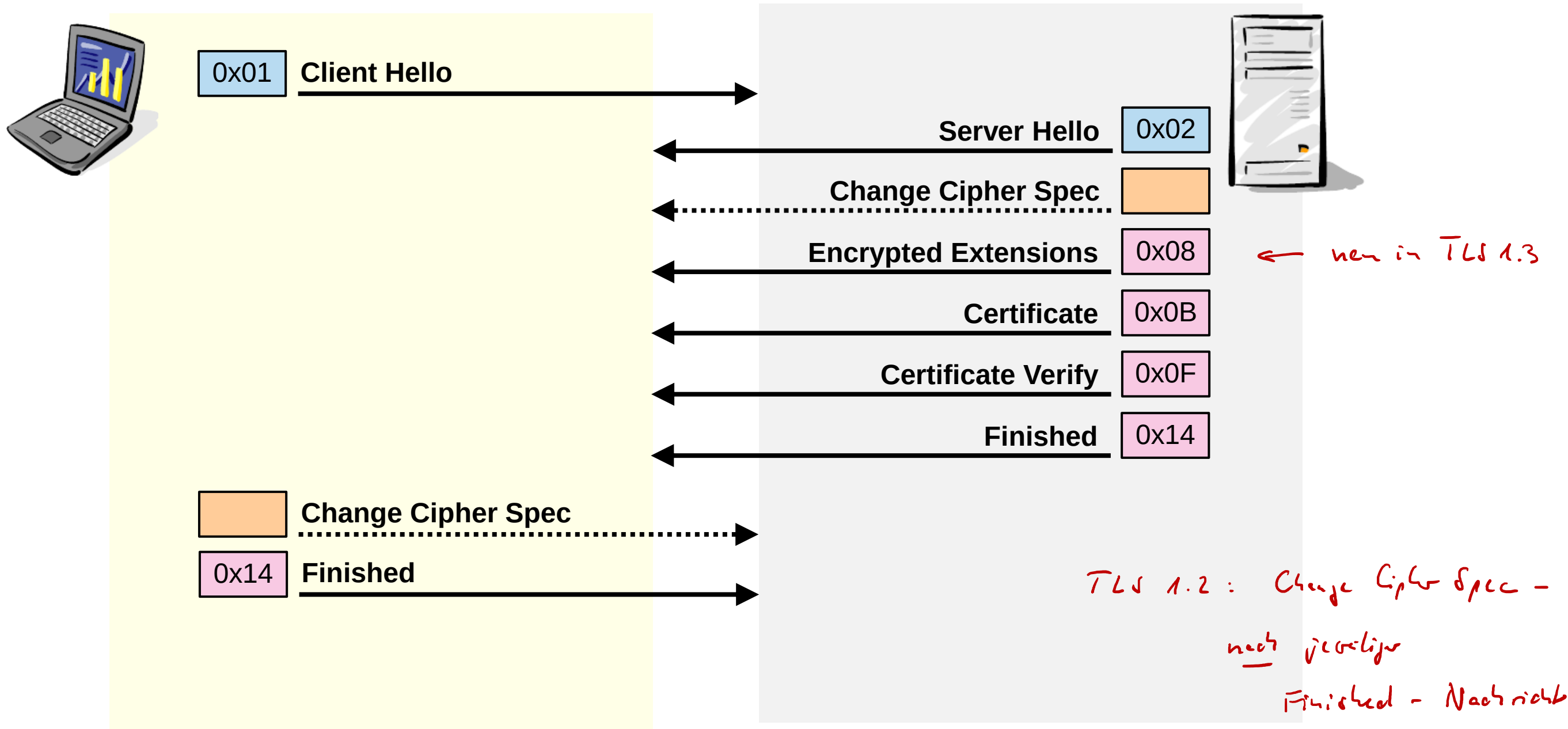
Seiteninformationen - <https://www.th-owl.de/iat/>

 Allgemein  Medien  Berechtigungen  Sicherheit

**Website-Identität**  
Website: [www.th-owl.de](https://www.th-owl.de)  
Besitzer: Diese Website stellt keine Informationen über den Besitzer zur Verfügung.  
Validiert von: Hellenic Academic and Research Institutions CA

**Datenschutz & Chronik**  
Habe ich diese Website früher schon einmal besucht? Nein  
Speichert diese Website Daten auf meinem Computer? Ja, 64,0 KB Website-Daten  
Habe ich Passwörter für diese Website gespeichert? Ja

**Technische Details**  
Verbindung verschlüsselt (TLS\_AES\_128\_GCM\_SHA256, 128-Bit-Schlüssel, TLS 1.3)  
Die Seite, die Sie ansehen, wurde verschlüsselt, bevor sie über das Internet übermittelt wurde.  
Verschlüsselung macht es für unberechtigte Personen schwierig, zwischen Computern übertragene Informationen anzusehen.  
Diese Website entspricht den Vorschriften für Zertifikat-Transparenz.



# Wireshark-Capture: TLS 1.3, Client Hello

\*WLAN

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telefonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No.       | Time | Source         | Destination     | Protocol | Leng  | Info                             |
|-----------|------|----------------|-----------------|----------|-------|----------------------------------|
| ... 34... |      | 192.168.114.56 | 193.174.116.151 | TLSv1.3  | 19... | Client Hello (SNI=www.th-owl.de) |

Cipher Suites Length: 32

✓ Cipher Suites (16 suites)

Cipher Suite: TLS\_AES\_128\_GCM\_SHA256 (0x1301)

Cipher Suite: TLS\_CHACHA20\_POLY1305\_SHA256 (0x1303)

Cipher Suite: TLS\_AES\_256\_GCM\_SHA384 (0x1302)

Cipher Suite: TLS\_ECDHE\_ECDSA\_WITH\_AES\_128\_GCM\_SHA256 (0xc02b)

Cipher Suite: TLS\_ECDHE\_RSA\_WITH\_AES\_128\_GCM\_SHA256 (0xc02f)

Cipher Suite: TLS\_ECDHE\_ECDSA\_WITH\_CHACHA20\_POLY1305\_SHA256 (0xc030)

Cipher Suite: TLS\_ECDHE\_RSA\_WITH\_CHACHA20\_POLY1305\_SHA256 (0xc031)

Cipher Suite: TLS\_ECDHE\_ECDSA\_WITH\_AES\_256\_GCM\_SHA384 (0xc02c)

Cipher Suite: TLS\_ECDHE\_RSA\_WITH\_AES\_256\_GCM\_SHA384 (0xc030)

Cipher Suite: TLS\_ECDHE\_ECDSA\_WITH\_AES\_256\_CBC\_SHA (0xc00a)

Cipher Suite: TLS\_ECDHE\_RSA\_WITH\_AES\_128\_CBC\_SHA (0xc013)

Cipher Suite: TLS\_ECDHE\_RSA\_WITH\_AES\_256\_CBC\_SHA (0xc014)

Cipher Suite: TLS\_RSA\_WITH\_AES\_128\_GCM\_SHA256 (0x009c)

Cipher Suite: TLS\_RSA\_WITH\_AES\_256\_GCM\_SHA384 (0x009d)

Cipher Suite: TLS\_RSA\_WITH\_AES\_128\_CBC\_SHA (0x002f)

Cipher Suite: TLS\_RSA\_WITH\_AES\_256\_CBC\_SHA (0x0035)

Compression Methods Length: 1

✓ Compression Methods (1 method)

Extensions Length: 1772

✓ Extension: server\_name (len=18) name=www.th-owl.de

✓ Extension: extended\_master\_secret (len=0)

✓ Extension: renegotiation\_info (len=1)

✓ Extension: supported\_groups (len=16)

✓ Extension: ec\_point\_formats (len=2)

✓ Extension: application\_layer\_protocol\_negotiation (len=14)

Type: application\_layer\_protocol\_negotiation (16)

Length: 14

ALPN Extension Length: 12

✓ ALPN Protocol

ALPN string length: 2

0080 e8 64 00 20 13 01 13 03 13 01  
0090 cc a8 c0 2c c0 30 c0 0a c0 13  
00a0 00 2f 00 35 01 00 06 ec 00 06  
00b0 0d 77 77 77 2e 74 68 2d 6f 77  
00c0 00 00 ff 01 00 01 00 00 0a 00  
00d0 1d 00 17 00 18 00 19 01 00 01  
00e0 00 00 10 00 0e 00 0c 02 68 3d  
00f0 31 2e 31 00 05 00 05 01 00 06  
0100 00 08 04 03 05 03 06 03 02 01  
0110 05 2f 05 2d 11 ec 04 c0 2f 01  
0120 61 5b e1 1b 5a 91 70 6a 0c 4d  
0130 9b d9 35 d6 e5 ca 23 9c 6a 27  
0140 25 2f 0c a4 41 7d 64 28 91 9a  
0150 1d 4f 95 36 d2 d7 87 9b d8 7f  
0160 b9 77 37 ac 6c 45 3e 5c 2b 4a  
0170 ac 37 d1 5c 51 d0 71 0c 30 0d  
0180 75 8b 00 c8 57 76 7e 65 69 9f  
0190 49 1a 68 d7 22 be 50 f4 1b c0  
01a0 a1 6b be 23 2a 37 24 58 db ba  
01b0 6e 2a c9 8b 80 18 a8 93 84 3f  
01c0 ee 34 45 d1 e0 ae fc c4 8e fc  
01d0 04 45 6e 1a 55 62 06 12 ac 3a  
01e0 9b 21 04 92 ef c5 81 39 a4 9a  
01f0 85 41 8c c6 63 0c 71 dc cb 4d  
0200 96 18 3d 9c 57 23 00 27 f1 8f  
0210 30 db 0a 4e 87 66 38 86 d6 c8  
0220 8f b0 19 b9 03 8f f1 09 12 ca  
0230 f0 9e 23 39 04 2a 56 7a ec d5  
0240 46 2a fc b4 16 c8 c1 e8 92 2f  
0250 1e 2b 6c c2 77 8d 87 f1 9d e5  
0260 e3 b3 ec eb 5b bf 61 cd 7b d5  
0270 17 8a 69 c0 84 5c 3a ba 94 4f  
0280 03 71 05 51 c2 2c eb 78 9d ae  
0290 46 42 56 44 27 83 b9 5e 11 e8  
02a0 69 c0 78 35 15 41 0b 22 28 0d  
02b0 19 14 43 53 50 4e c3 49 92 1e

Cipher Suite (tls.handshake.ciphersuite), 2 Bytes

Pakete: 2541 - Angezeigt: 265 (10.4%) - Verworfen: 0 (0.0%) - Profil: Default

# Wireshark-Capture: TLS 1.3, Client Hello

\*WLAN

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telefonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No.       | Time | Source         | Destination     | Protocol | Leng  | Info                             |
|-----------|------|----------------|-----------------|----------|-------|----------------------------------|
| ... 34... |      | 192.168.114.56 | 193.174.116.151 | TLSv1.3  | 19... | Client Hello (SNI=www.th-owl.de) |

Extensions Length: 1772

- > Extension: server\_name (len=18) name=www.th-owl.de
- > Extension: extended\_master\_secret (len=0)
- > Extension: renegotiation\_info (len=1)
- ✓ Extension: supported\_groups (len=16)
  - Type: supported\_groups (10)
  - Length: 16
  - Supported Groups List Length: 14
  - ✓ Supported Groups (7 groups)
    - Supported Group: X25519MLKEM768 (0x11ec)
    - Supported Group: x25519 (0x001d)
    - Supported Group: secp256r1 (0x0017)
    - Supported Group: secp384r1 (0x0018)
    - Supported Group: secp521r1 (0x0019)
    - Supported Group: ffdhe2048 (0x0100)
    - Supported Group: ffdhe3072 (0x0101)
- > Extension: ec\_point\_formats (len=2)
- ✓ Extension: application\_layer\_protocol\_negotiation (len=14)
  - Type: application\_layer\_protocol\_negotiation (16)
  - Length: 14
  - ALPN Extension Length: 12
  - ✓ ALPN Protocol
    - ALPN string length: 2
    - ALPN Next Protocol: h2
    - ALPN string length: 8
    - ALPN Next Protocol: http/1.1
- > Extension: status\_request (len=5)
- > Extension: delegated\_credentials (len=10)
- > Extension: signed\_certificate\_timestamp (len=0)
- ✓ Extension: key\_share (len=1327) X25519MLKEM768, x25519, secp256r1
  - Type: key\_share (51)
  - Length: 1327

ff 01 00 01 00 00 0a 00 10 00 0e 11  
17 00 18 00 19 01 00 01 01 00 0b 00  
10 00 0e 00 0c 02 68 32 08 68 74 74  
31 00 05 00 05 01 00 00 00 00 00 22  
04 03 05 03 06 03 02 03 00 12 00 00  
05 2d 11 ec 04 c0 2f 0b 0c 35 6c ab  
e1 1b 5a 91 70 6a 0c 43 79 a5 0a 29  
35 d6 e5 ca 23 9c 6a 27 40 1c 6f 51  
0c a4 41 7d 64 28 91 9a 3f 7d 56 5f  
95 36 d2 d7 87 9b d8 75 c7 aa cd b0  
37 ac 6c 45 3e 5c 2b 4a 91 09 55 08  
d1 5c 51 d0 71 0c 30 02 bf b1 e7 00  
00 c8 57 76 7e 65 69 96 62 4b 63 e0  
68 d7 22 be 50 f4 1b cd aa c4 cc 42  
be 23 2a 37 24 58 db ba ca e4 31 07  
c9 8b 80 18 a8 93 84 35 06 66 87 c9  
45 d1 e0 ae fc c4 8e fc 18 9a 3a 57  
6e 1a 55 62 06 12 ac 3a 74 67 22 84  
04 92 ef c5 81 39 a4 94 69 d2 b0 ef  
8c c6 63 0c 71 dc cb 4f d4 c2 d3 fb  
3d 9c 57 23 00 27 f1 88 b3 c2 f3 cb  
0a 4e 87 66 38 86 d6 c8 45 f1 a6 7b  
19 b9 03 8f f1 09 12 ca ac 26 4b 94  
23 39 04 2a 56 7a ec d5 1e 27 12 1c  
fc b4 16 c8 c1 e8 92 2f c4 eb 00 23  
6c c2 77 8d 87 f1 9d e3 91 6a 75 73  
ec eb 5b bf 61 cd 7b d9 0b 08 e3 c6  
69 c0 84 5c 3a ba 94 45 f1 f3 5b 59  
05 51 c2 2c eb 78 9d ae 6a 32 a0 18  
56 44 27 83 b9 5e 11 e8 91 aa fc 74  
78 35 15 41 0b 22 28 03 27 fb 25 28  
43 53 50 4e c3 49 92 1e 84 1d 6a 9b  
5a 70 a9 32 61 a0 58 c4 ac 02 21 3f  
11 a6 a0 aa 50 20 3b 5d 17 f4 6c 36  
0a e6 c8 01 d4 6c a3 62 85 86 07 2c  
c2 87 12 2c c3 1b 61 9b 50 9f 85 95

EC  
Finite Fields (nur für TLS 1.2)

Hello extension (tls.handshake.extension), 20 Bytes

Pakete: 2541 - Angezeigt: 265 (10.4%) - Verworfen: 0 (0.0%) - Profil: Default



# Wireshark-Capture: TLS 1.3, Client Hello

\*WLAN

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No.       | Time | Source         | Destination     | Protocol | Leng  | Info                             |
|-----------|------|----------------|-----------------|----------|-------|----------------------------------|
| ... 34... |      | 192.168.114.56 | 193.174.116.151 | TLSv1.3  | 19... | Client Hello (SNI=www.th-owl.de) |

> Extension: delegated\_credentials (len=10)

> Extension: signed\_certificate\_timestamp (len=0)

> Extension: key\_share (len=1327) X25519MLKEM768, x25519, secp256r1

Type: key\_share (51)  
Length: 1327

> Key Share extension

Client Key Share Length: 1325

① > Key Share Entry: Group: X25519MLKEM768, Key Exchange length: 1216

Group: X25519MLKEM768 (4588)  
Key Exchange Length: 1216  
Key Exchange [...]: 2f0b0c356cab80d6615be11b5a91706a0c4379a50a2920879bd935d6e5ca239c6a27401c6f5149a5252f0ca4417d6428

② > Key Share Entry: Group: x25519, Key Exchange length: 32

Group: x25519 (29)  
Key Exchange Length: 32  
Key Exchange: 0a68ff27ed34165d4af0e3802a1cb672b8d77b702fe4add8328b876c1c2f5f71

③ > Key Share Entry: Group: secp256r1, Key Exchange length: 65

Group: secp256r1 (23)  
Key Exchange Length: 65  
Key Exchange: 04575d5cc8a98371a23759c46800faa8bc6049a7a501b8252786b1bb36399f561c3dc4c3f4433fa71287fa1cb433241d1452

> Extension: supported\_versions (len=5) TLS 1.3, TLS 1.2

> Extension: signature\_algorithms (len=24)

Type: signature\_algorithms (13)  
Length: 24  
Signature Hash Algorithms Length: 22

> Signature Hash Algorithms (11 algorithms)

> Extension: record\_size\_limit (len=2)

> Extension: compress\_certificate (len=7)

> Extension: encrypted\_client\_hello (len=281)

[JA4: t13d1615h2\_86a278354501\_a54fffd0eb61]  
[JA4\_r: t13d1615h2\_002f,0035,009c,009d,1301,1302,1303,c00a,c013,c014,c02b,c02c,c02f,c030,cca8,cca9\_0005,000a,000b,000d,  
[JA3 Fullstring: 771,4865-4867-4866-49195-49199-52393-52392-49196-49200-49162-49171-49172-156-157-47-53,0-23-65281-10-1  
[JA3: b074ad2e08e2809192a32ad8042d1940]

|    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|
| 04 | 03 | 05 | 03 | 06 | 03 | 02 | 03 | 00 | 12 | 00 | 00 |
| 05 | 2d | 11 | ec | 04 | c0 | 2f | 0b | 0c | 35 | 6c | ab |
| e1 | 1b | 5a | 91 | 70 | 6a | 0c | 43 | 79 | a5 | 0a | 29 |
| 35 | d6 | e5 | ca | 23 | 9c | 6a | 27 | 40 | 1c | 6f | 51 |
| 0c | a4 | 41 | 7d | 64 | 28 | 91 | 9a | 3f | 7d | 56 | 5f |
| 95 | 36 | d2 | d7 | 87 | 9b | d8 | 75 | c7 | aa | cd | b0 |
| 37 | ac | 6c | 45 | 3e | 5c | 2b | 4a | 91 | 09 | 55 | 08 |
| d1 | 5c | 51 | d0 | 71 | 0c | 30 | 02 | bf | b1 | e7 | 00 |
| 00 | c8 | 57 | 76 | 7e | 65 | 69 | 96 | 62 | 4b | 63 | e0 |
| 68 | d7 | 22 | be | 50 | f4 | 1b | cd | aa | c4 | cc | 42 |
| be | 23 | 2a | 37 | 24 | 58 | db | ba | ca | e4 | 31 | 07 |
| c9 | 8b | 80 | 18 | a8 | 93 | 84 | 35 | 06 | 66 | 87 | c9 |
| 45 | d1 | e0 | ae | fc | c4 | 8e | fc | 18 | 9a | 3a | 57 |
| 6e | 1a | 55 | 62 | 06 | 12 | ac | 3a | 74 | 67 | 22 | 84 |
| 04 | 92 | ef | c5 | 81 | 39 | a4 | 94 | 69 | d2 | b0 | ef |
| 8c | c6 | 63 | 0c | 71 | dc | cb | 4f | d4 | c2 | d3 | fb |
| 3d | 9c | 57 | 23 | 00 | 27 | f1 | 88 | b3 | c2 | f3 | cb |
| 0a | 4e | 87 | 66 | 38 | 86 | d6 | c8 | 45 | f1 | a6 | 7b |
| 19 | b9 | 03 | 8f | f1 | 09 | 12 | ca | ac | 26 | 4b | 94 |
| 23 | 39 | 04 | 2a | 56 | 7a | ec | d5 | 1e | 27 | 12 | 1c |
| fc | b4 | 16 | c8 | c1 | e8 | 92 | 2f | c4 | eb | 00 | 23 |
| 6c | c2 | 77 | 8d | 87 | f1 | 9d | e3 | 91 | 6a | 75 | 73 |
| ec | eb | 5b | bf | 61 | cd | 7b | d9 | 0b | 08 | e3 | c6 |
| 69 | c0 | 84 | 5c | 3a | ba | 94 | 45 | f1 | f3 | 5b | 59 |
| 05 | 51 | c2 | 2c | eb | 78 | 9d | ae | 6a | 32 | a0 | 18 |
| 56 | 44 | 27 | 83 | b9 | 5e | 11 | e8 | 91 | aa | fc | 74 |
| 78 | 35 | 15 | 41 | 0b | 22 | 28 | 03 | 27 | fb | 25 | 28 |
| 43 | 53 | 50 | 4e | c3 | 49 | 92 | 1e | 84 | 1d | 6a | 9b |
| 5a | 70 | a9 | 32 | 61 | a0 | 58 | c4 | ac | 02 | 21 | 3f |
| 11 | a6 | a0 | aa | 50 | 20 | 3b | 5d | 17 | f4 | 6c | 36 |
| 0a | e6 | c8 | 01 | d4 | 6c | a3 | 62 | 85 | 86 | 07 | 2c |
| c2 | 87 | 12 | 2c | c3 | 1b | 61 | 9b | 50 | 9f | 85 | 95 |
| c2 | bb | b3 | d8 | 96 | 8f | 44 | 63 | 48 | a1 | 2f | 9f |
| 1e | 85 | 75 | 5a | 2c | 64 | 7b | d3 | 7c | 59 | fc | 6a |
| 35 | c9 | c9 | fc | fb | c5 | 58 | f8 | 25 | 94 | 79 | 46 |
| 48 | 63 | 48 | 99 | ae | 69 | 61 | 7a | 12 | 17 | b6 | 63 |

Public key

Hello extension (tls.handshake.extension), 1.331 Bytes

Pakete: 2541 · Angezeigt: 265 (10.4%) · Verworfen: 0 (0.0%)

Profil: Default

# Wireshark-Capture: TLS 1.3, Server Hello

\*WLAN

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time  | Source          | Destination     | Protocol | Leng  | Info                                                 |
|-----|-------|-----------------|-----------------|----------|-------|------------------------------------------------------|
| ... | 34... | 192.168.114.56  | 193.174.116.151 | TLSv1.3  | 19... | Client Hello (SNI=www.th-owl.de)                     |
| ... | 34... | 193.174.116.151 | 192.168.114.56  | TLSv1.3  | 13... | Server Hello, Change Cipher Spec, Application Data   |
| ... | 34... | 193.174.116.151 | 192.168.114.56  | TLSv1.3  | 12... | Application Data, Application Data, Application Data |
| ... | 34... | 192.168.114.56  | 193.174.116.151 | TLSv1.3  | 118   | Change Cipher Spec, Application Data                 |

Random: 9f7b8d672c0bc22fecb0ff5011f38b72997b7caa8bb83482e73e1ebc29e70043  
Session ID Length: 32  
Session ID: e0799c39c3356bf1f8451c1d7b8fb8bfe75165fc37e7a2a40899893d02e8e864  
Cipher Suite: TLS\_AES\_128\_GCM\_SHA256 (0x1301)  
Compression Method: null (0)  
Extensions Length: 79

Extension: key\_share (len=69) secp256r1  
Type: key\_share (51)  
Length: 69

Key Share extension  
Key Share Entry: Group: secp256r1, Key Exchange length: 65  
Group: secp256r1 (23)  
Key Exchange Length: 65  
Key Exchange: 047fcf2044c553b765a4f47324637523d7dd5c85fccb51d340b4686ad90cbe7488ac0616c416d051cf45b829ff4b6d91d0a3

Extension: supported\_versions (len=2) TLS 1.3  
[JA3S Fullstring: 771,4865,51-43]  
[JA3S: eb1d94daa7e0344597e756a1fb6e7054]

TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec  
Content Type: Change Cipher Spec (20)  
Version: TLS 1.2 (0x0303)  
Length: 1  
Change Cipher Spec Message

TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol  
Opaque Type: Application Data (23)  
Version: TLS 1.2 (0x0303)  
Length: 23  
Encrypted Application Data: a886e704f2130673d7f62aa7dcb545a51c7da8b561b70b  
[Application Data Protocol: Hypertext Transfer Protocol]  
TLS segment data (1056 bytes)

0080 e8 64 13 01 00 00 4f 00 33 00  
0090 7f cf 20 44 c5 53 b7 65 a4 f4  
00a0 dd 5c 85 fc cb 51 d3 40 b4 68  
00b0 ac 06 16 c4 16 d0 51 cf 45 b8  
00c0 a3 bf 3b c8 81 26 70 f5 95 46  
00d0 00 2b 00 02 03 04 14 03 03 06  
00e0 17 a8 86 e7 04 f2 13 06 73 d7  
00f0 a5 1c 7d a8 b5 61 b7 0b 17 03  
0100 16 f6 7a 3c 8c 38 bf 23 1c 99  
0110 0f a2 7a 56 85 98 72 0e 63 1f  
0120 b0 50 04 4f 45 93 4f 9c 9b a3  
0130 54 12 27 2e c3 87 ad 21 24 c7  
0140 96 c2 13 73 38 98 f2 d1 0f 81  
0150 c7 da 90 8d dc 0f 6d 81 1e 54  
0160 41 fc 9a eb bf 57 7f 83 57 8c  
0170 a7 69 7f 5d fa 66 14 24 ca 8c  
0180 26 7f 94 a2 51 86 90 32 0b fe  
0190 0a 5b 16 ef 7d dc b2 5a 8e 4f  
01a0 08 ff 85 e1 19 a6 8e ab 90 2f  
01b0 98 0e 3d 6c a9 34 bb f1 d3 58  
01c0 73 2e 08 6f 42 0e 5d 47 76 1e  
01d0 8f fb 85 6b 3d 10 9b 5d f0 3f  
01e0 f5 35 35 42 17 90 96 41 37 5c  
01f0 4b 4c 01 78 b3 17 bb 0c f0 fc  
0200 c2 c6 21 87 1b cc 95 4a fb 8e  
0210 58 03 fa 91 ea e7 9a 9c 73 87  
0220 e7 f5 c2 3e 5e 1d d8 ea 88 fe  
0230 e3 a4 d9 05 4d 23 24 ef f1 03  
0240 14 ed de 83 3c e5 93 de 01 57  
0250 a4 ba 79 9b 03 04 59 a9 05 64  
0260 81 d7 d9 2f e2 f0 ec 32 d0 e4  
0270 00 5b 17 e5 aa 1f 40 7e 29 6f  
0280 1e 7e f7 87 8c 86 7c dc c5 e8  
0290 27 27 27 27 27 27 27 27 27 27

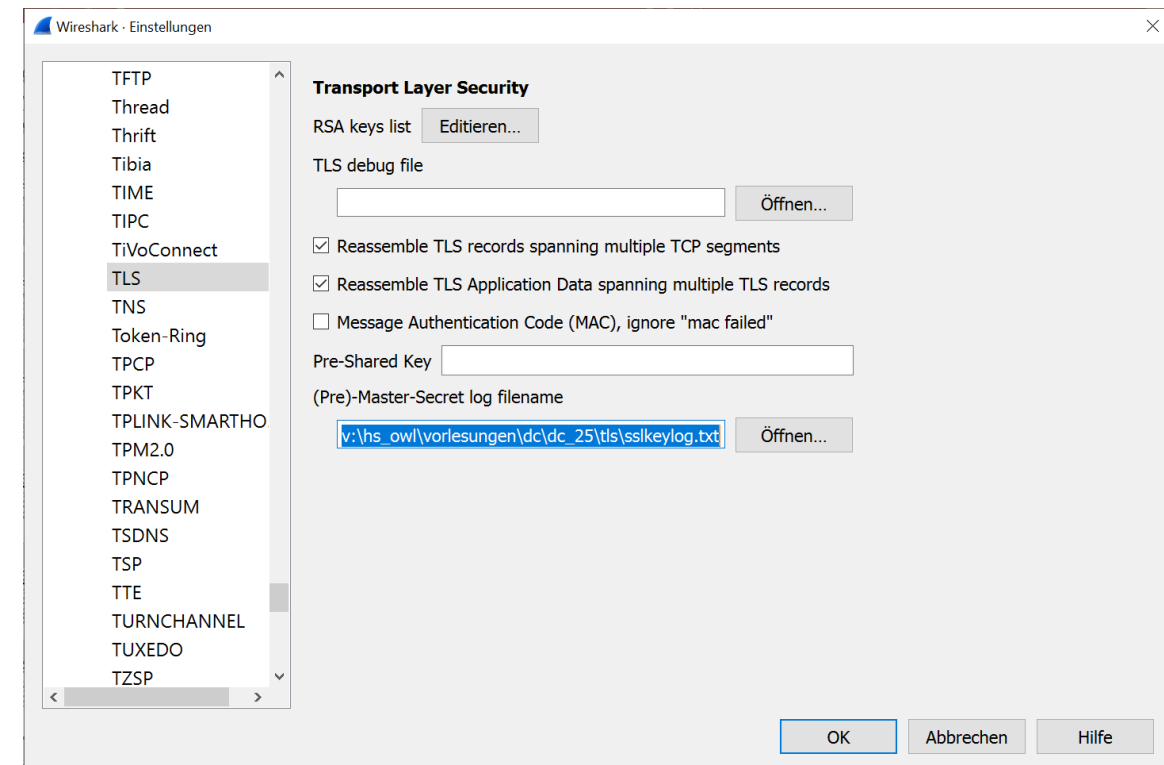
③ am Client Hello  
public key

Hello extension (tls.handshake.extension), 73 Bytes

Pakete: 2541 - Angezeigt: 265 (10.4%) - Verworfen: 0 (0.0%)

Profil: Default

- Aktivierung der Schlüsselspeicherung in einer **Key-Log-Datei** durch einen geeigneten Browser ([NSS Key Log Format](#)):
  - Nach Setzen eines Dateinamens zur Speicherung der TLS-Sitzungsschlüssel (Key-Log-Datei) in der Umgebungsvariablen **SSLKEYLOGFILE**, werden vom Browser die Schlüssel in dieser Key-Log-Datei automatisch abgespeichert.
- **Konfiguration von Wireshark:**
  - -> Bearbeiten
  - -> Einstellungen...
  - -> Protocols -> TLS
  - Angabe der Key-Log-Datei unter **(Pre)-Master-Secret log filename**



dc\_260618.pcapng

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time   | Source        | Source Port | Destination     | Protocol | Length | Info                                                   |
|-----|--------|---------------|-------------|-----------------|----------|--------|--------------------------------------------------------|
| 290 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 1940   | Client Hello (SNI=www.th-owl.de)                       |
| 292 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1304   | Server Hello, Change Cipher Spec, Encrypted Extensions |
| 296 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1200   | Certificate, Certificate Verify, Finished              |
| 298 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 118    | Change Cipher Spec, Finished                           |
| 311 | 35.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 802    |                                                        |

Version: TLS 1.2 (0x0303)  
Length: 155  
> Handshake Protocol: Server Hello

✓ TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec  
Content Type: Change Cipher Spec (20)  
Version: TLS 1.2 (0x0303)  
Length: 1  
Change Cipher Spec Message

✓ TLSv1.3 Record Layer: Handshake Protocol: Encrypted Extensions  
Opaque Type: Application Data (23)  
Version: TLS 1.2 (0x0303)  
Length: 23  
[Content Type: Handshake (22)]  
[Sequence Number: 0]

✓ Handshake Protocol: Encrypted Extensions  
Handshake Type: Encrypted Extensions (8)  
Length: 2  
Extensions Length: 0  
TLS segment data (1056 bytes)

0000 08 00 00 02 00 00

Packet (1304 bytes)Decrypted TLS (6 bytes)

Pakete: 2541 · Angezeigt: 192 (7.6%) | Profil: Default

Handshake protocol message (tls.handshake), 6 Bytes



dc\_260618.pcapng

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time   | Source        | Source Port | Destination     | Protocol | Length | Info                                                   |
|-----|--------|---------------|-------------|-----------------|----------|--------|--------------------------------------------------------|
| 290 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 1940   | Client Hello (SNI=www.th-owl.de)                       |
| 292 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1304   | Server Hello, Change Cipher Spec, Encrypted Extensions |
| 296 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1200   | Certificate, Certificate Verify, Finished              |
| 298 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 118    | Change Cipher Spec, Finished                           |
| 311 | 35.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 802    |                                                        |

[Content Type: Handshake (22)]  
[Sequence Number: 1]  
Handshake Protocol: Certificate  
Handshake Type: Certificate (11)  
Length: 5326  
Certificate Request Context Length: 0  
Certificates Length: 5322  
Certificates (5322 bytes)  
Certificate Length: 2030  
Certificate [...]: 308207ea30820652a00302010202103e2404adec38ea05932ec6d52f7e1196300d06092a864886f70d0  
Extensions Length: 0  
Certificate Length: 1545  
Certificate [...]: 30820605308203eda003020102021014d57bf3692228219a5567fa91651b22300d06092a864886f70d0  
Extensions Length: 0  
Certificate Length: 1732  
Certificate [...]: 308206c0308204a8a00302010202102a6086d4d4de45c95e4b98fbbf2fbf26300d06092a864886f70d0  
Extensions Length: 0  
Transport Layer Security  
[Stream index: 11]

0000 0b 00 14 ce 00 00 14 ca 00 07  
0010 82 06 52 a0 03 02 01 02 02 10  
0020 ea 05 93 2e c6 d5 2f 7e 11 96  
0030 48 86 f7 0d 01 01 0b 05 00 30  
0040 03 55 04 06 13 02 47 52 31 37  
0050 0a 0c 2e 48 65 6c 6c 65 6e 69  
0060 65 6d 69 63 20 61 6e 64 20 52  
0070 68 20 49 6e 73 74 69 74 75 74  
0080 41 31 18 30 16 06 03 55 04 03  
0090 54 20 54 4c 53 20 52 53 41 20  
00a0 35 31 30 32 39 31 32 35 37 35  
00b0 31 30 32 39 31 32 35 37 35 35  
00c0 09 06 03 55 04 06 13 02 44 45  
00d0 55 04 08 0c 13 4e 6f 72 64 72  
00e0 65 73 74 66 61 6c 65 6e 31 31  
00f0 0a 0c 28 54 65 63 68 6e 69 73  
0100 63 68 73 63 68 75 6c 65 20 4f  
0110 66 61 6c 65 6e 2d 4c 69 70 70  
0120 03 55 04 03 0c 09 74 68 2d 6f  
0130 82 02 22 30 0d 06 09 2a 86 48

Decrypted TCP (5352 bytes)Decrypted TLS (5330 bytes)

Pakete: 2541 · Angezeigt: 192 (7.6%)Profil: Default

18.06.2026

Datensicherheit – 2025 – Prof. S. Heiss

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Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time      | Source          | Source Port | Destination     | Protocol | Length | Info                                                    |
|-----|-----------|-----------------|-------------|-----------------|----------|--------|---------------------------------------------------------|
| 290 | 34.000000 | 192.168.114.100 | 56709       | 193.174.116.151 | TLSv1.3  | 1940   | Client Hello (SNI=www.th-owl.de)                        |
| 292 | 34.000000 | 193.174.116.151 | 443         | 192.168.114.100 | TLSv1.3  | 1304   | Server Hello, Change Cipher Spec, Encrypted Extensions  |
| 296 | 34.000000 | 193.174.116.151 | 443         | 192.168.114.100 | TLSv1.3  | 1200   | Certificate, Certificate Verify, Finished               |
| 298 | 34.000000 | 192.168.114.100 | 56709       | 193.174.116.151 | TLSv1.3  | 118    | Change Cipher Spec, Finished                            |
| 311 | 35.000000 | 193.174.116.151 | 443         | 192.168.114.100 | TLSv1.3  | 802    |                                                         |
| 432 | 35.000000 | 192.168.114.100 | 56709       | 193.174.116.151 | TLSv1.3  | 78     | Alert (Level: Warning, Description: Close Notification) |

[Content Type: Handshake (22)]  
[Sequence Number: 1]  
✓ Handshake Protocol: Certificate  
Handshake Type: Certificate (11)  
Length: 5326  
Certificate Request Context Length: 0  
Certificates Length: 5322  
✓ Certificates (5322 bytes)  
Certificate Length: 2030  
✓ Certificate [...]: 308207ea30820652a00302010202103e2404adec38ea05932ec6d52f7e1196300d06092a864886f70d01010b0500306031  
✓ signedCertificate  
version: v3 (2)  
serialNumber: 0x3e2404adec38ea05932ec6d52f7e1196  
> signature (sha256WithRSAEncryption)  
> issuer: rdnSequence (0)  
> validity  
✓ subject: rdnSequence (0)  
> rdnSequence: 4 items (id-at-commonName=th-owl.de,id-at-organizationName=Technische Hochschule Ostwestfalen-Lippe)  
> subjectPublicKeyInfo  
> extensions: 9 items  
> algorithmIdentifier (sha256WithRSAEncryption)  
Padding: 0

00b0 31 30 32 39 31 32 35 37 35 35 5a 30  
00c0 09 06 03 55 04 06 13 02 44 45 31 1c  
00d0 55 04 08 0c 13 4e 6f 72 64 72 68 65  
00e0 65 73 74 66 61 6c 65 6e 31 31 30 2f  
00f0 0a 0c 28 54 65 63 68 6e 69 73 63 68  
0100 63 68 73 63 68 75 6c 65 20 4f 73 74  
0110 66 61 6c 65 6e 2d 4c 69 70 70 65 31  
0120 03 55 04 03 0c 09 74 68 2d 6f 77 6c  
0130 82 02 22 30 0d 06 09 2a 86 48 86 f7  
0140 05 00 03 82 02 0f 00 30 82 02 0a 02  
0150 90 cb 6d 46 1e 43 10 ff 38 b1 20 72  
0160 98 83 98 5f ca a8 aa 7a 5e 2c 40 2f  
0170 95 d2 b8 45 b0 11 15 f7 9f 89 10 81  
0180 1b 21 0a e1 6f 16 3d 4c ee bb de 22  
0190 01 44 94 0b ae b6 68 5a a0 e0 20 3c  
01a0 63 a2 60 fc 3e e4 1f 01 41 9a 47 44  
01b0 59 62 6d d1 39 d2 a6 49 ec 29 9a d2  
01c0 e3 6d 46 2c 70 b8 24 a1 62 29 51 12  
01d0 78 4f 9a 7c f0 a2 10 d8 f1 24 fe 8f  
01e0 3b 2a 67 3f de 7c 96 f5 8c bb c0 dd  
01f0 a2 a6 9a 00 00 6f 41 40 55 ac 4d 1d  
0200 76 f2 ff 42 d5 fc 96 14 ef 0e a7 cf  
0210 8a ee ef 45 d4 f9 cc 66 38 58 31 37  
0220 27 3f d4 67 af 94 25 ee 7b c5 8c c6

Decrypted TCP (5352 bytes) Decrypted TLS (5330 bytes)

Pakete: 2541 · Angezeigt: 192 (7.6%) Profil: Default

# Wireshark-Capture: TLS 1.3, Certificate Verify, Finished

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Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time   | Source        | Source Port | Destination     | Protocol | Length | Info                                                   |
|-----|--------|---------------|-------------|-----------------|----------|--------|--------------------------------------------------------|
| 290 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 1940   | Client Hello (SNI=www.th-owl.de)                       |
| 292 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1304   | Server Hello, Change Cipher Spec, Encrypted Extensions |
| 296 | 34.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 1200   | Certificate, Certificate Verify, Finished              |
| 298 | 34.... | 192.168.11... | 56709       | 193.174.116.151 | TLSv1.3  | 118    | Change Cipher Spec, Finished                           |
| 311 | 35.... | 193.174.11... | 443         | 192.168.114.56  | TLSv1.3  | 802    |                                                        |

[Sequence Number: 2]

- Handshake Protocol: Certificate Verify
  - Handshake Type: Certificate Verify (15)
  - Length: 516
  - Signature Algorithm: rsa\_pss\_rsae\_sha256 (0x0804)
    - Signature Hash Algorithm Hash: Unknown (8)
    - Signature Hash Algorithm Signature: Unknown (4)
    - Signature length: 512
    - Signature [...]: 11ed7d296d609aa7a86d863082e240f1d1f7ceb61bc9099eccdd3fb800200a88249d47aec06035ed111577d
- TLSv1.3 Record Layer: Handshake Protocol: Finished
  - Opaque Type: Application Data (23)
  - Version: TLS 1.2 (0x0303)
  - Length: 53
  - [Content Type: Handshake (22)]
  - [Sequence Number: 3]
  - Handshake Protocol: Finished
    - Handshake Type: Finished (20)
    - Length: 32
    - Verify Data: 554a764bd2f0da857ceb1df0f3c8a662637553ff10274a60a535fae906b3d642

0000 0f 00 02 04 08 04 02 00 11 ed  
0010 a8 6d 86 30 82 e2 40 f1 d1 f7  
0020 cc dd 3f b8 00 20 0a 88 24 9d  
0030 11 15 77 dc 5e 6d 87 02 44 12  
0040 72 4d 7d 87 e5 07 9c 89 22 20  
0050 5d 9a ad 0c 65 e4 3b 78 be b4  
0060 3f d6 00 11 d3 4f 00 27 03 73  
0070 43 2a fe da 10 8f 92 c0 be 71  
0080 95 5d 25 30 1a e9 98 05 ac 0c  
0090 97 20 76 09 ee 11 21 bb 36 24  
00a0 99 f7 ca cd 86 0d a8 52 0f 55  
00b0 83 d3 e7 d6 10 7a cc 97 1a 3a  
00c0 6f 51 0e b5 06 9b 80 9c fa 93  
00d0 39 7d f9 da 73 c2 c9 d1 26 8a  
00e0 a1 ce 00 37 4a 12 d8 87 cd a2  
00f0 e2 a3 86 6f d4 ff 3c 5e 38 6e  
0100 ab 5e 73 59 49 98 4c 3b 05 ed  
0110 d0 53 eb 30 e6 e0 c3 e3 5f b2  
0120 62 08 d4 0a 4f c6 ed 45 b0 8b  
0130 4c ce 26 eb 7c 0c db 4b 21 f3

Decrypted TLS (520 bytes) Decrypted TLS (36 bytes)

Type of handshake message (tls.handshake.type), 1 Byte

Pakete: 2541 · Angezeigt: 192 (7.6%) Profil: Default

# Wireshark-Capture: TLS 1.3, Change Cipher Spec, Finished (Client)

dc\_260618.pcapng

Datei Bearbeiten Ansicht Navigation Aufzeichnen Analyse Statistiken Telephonie Wireless Tools Hilfe

\_ws.col.protocol == "TLSv1.3"

| No. | Time      | Source         | Source Port | Destination     | Protocol | Length | Info                                                   |
|-----|-----------|----------------|-------------|-----------------|----------|--------|--------------------------------------------------------|
| 290 | 34.000000 | 192.168.114.56 | 56709       | 193.174.116.151 | TLSv1.3  | 1940   | Client Hello (SNI=www.th-owl.de)                       |
| 292 | 34.000000 | 193.174.114.56 | 443         | 192.168.114.56  | TLSv1.3  | 1304   | Server Hello, Change Cipher Spec, Encrypted Extensions |
| 296 | 34.000000 | 193.174.114.56 | 443         | 192.168.114.56  | TLSv1.3  | 1200   | Certificate, Certificate Verify, Finished              |
| 298 | 34.000000 | 192.168.114.56 | 56709       | 193.174.116.151 | TLSv1.3  | 118    | Change Cipher Spec, Finished                           |
| 311 | 35.000000 | 193.174.114.56 | 443         | 192.168.114.56  | TLSv1.3  | 802    |                                                        |

> Internet Protocol Version 4, Src: 192.168.114.56, Dst: 193.174.116.151  
> Transmission Control Protocol  
v Transport Layer Security  
[Stream index: 11]  
v TLSv1.3 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec  
Content Type: Change Cipher Spec (20)  
Version: TLS 1.2 (0x0303)  
Length: 1  
Change Cipher Spec Message  
v TLSv1.3 Record Layer: Handshake Protocol: Finished  
Opaque Type: Application Data (23)  
Version: TLS 1.2 (0x0303)  
Length: 53  
[Content Type: Handshake (22)]  
[Sequence Number: 0]  
v Handshake Protocol: Finished  
Handshake Type: Finished (20)  
Length: 32  
Verify Data: 1a941f19731c502364afd6c3e78bd7bdeb0b7830ee1b94514419ecda9b830b3

0000 aa 01 00 01 26 03 98 af 65 e6 8a  
0010 00 68 6a b3 40 00 80 06 00 00 c0  
0020 74 97 dd 85 01 bb b8 48 1f 09 03  
0030 00 ff 69 81 00 00 14 03 03 00 01  
0040 35 f4 c2 78 1a a7 0b 96 0a 28 2d  
0050 8a 60 96 77 20 ec ef 92 92 e8 c5  
0060 52 60 96 b7 a6 e1 87 52 04 0b 55  
0070 70 01 c8 3a bf 81

Packet (118 bytes) Decrypted TLS (36 bytes)  
Pakete: 2541 · Angezeigt: 192 (7.6%) Profil: Default